**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 054 · Boolean Conditions in Code

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What are 17 DIV 5 and 17 MOD 5?

Q2. What happens in the fetch stage?
A. Work out its meaning
B. Get the next instruction
C. Carry out the instruction
D. Save a backup copy

Q3. “Execution only copies an instruction into memory without performing it.” Correct this. Explain why.

Q4. A CPU compares scores and adds a bonus. Correct both claims. Explain why. (a) The Control Unit stores documents forever. (b) The ALU stores the next instruction's address.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Boolean — A Boolean value represents true or false.

AND — An AND condition is true only when both component conditions are true.

OR — An OR condition is true if at least one condition is true.

NOT — NOT reverses a Boolean value, changing true to false or false to true.

Algorithm — An ordered set of instructions to solve a problem.

Trace — Following instructions and recording changes to values or output.

## New learning

What is Boolean?: Boolean — A Boolean value represents true or false.

In this example: A swimming-pool entry policy uses and; An AND condition is true only when both component conditions are true.

Follow the effect: An OR condition is true if at least one condition is true. Explain how this changes the eventual program result.

Check understanding: NOT is used for a specific reason: NOT reverses a Boolean value, changing true to false or false to true.

## Worked example

A swimming-pool entry policy is being written. Explain how boolean and and are used to make the program work. [4 marks]

1. We are designing a swimming-pool entry policy. Identify which data and decisions are essential.

2. Boolean [1] — A Boolean value represents true or false. [1].

3. AND [1] — An AND condition is true only when both component conditions are true. [1].

4. Two linked points for each technique give four mark points.

# We Do · Practise with guidance

## Guided 1

In a club membership check, explain how boolean helps the program. [2 marks]

Worked method: Boolean [1] — A Boolean value represents true or false. [1].

## Guided 2





BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. DIV gives integer quotient 3; MOD gives remainder 2.

Q2. B — Get the next instruction

Q3. The operation specified by the decoded instruction is carried out.

Q4. (a) The Control Unit decodes instructions and controls CPU operations. (b) The ALU handles arithmetic and logic. Registers store data or addresses briefly.

### I Do / We Do pages · Guided practice

Q1. Boolean [1] — A Boolean value represents true or false. [1].

Q2. AND [1] — An AND condition is true only when both component conditions are true. [1].

Q3. OR [1] — An OR condition is true if at least one condition is true. [1]. NOT [1] — NOT reverses a Boolean value, changing true to false or false to true. [1].

### You Do pages · Code First

```
hasPassword = true
hasSecondFactor = false
print(hasPassword AND hasSecondFactor)
```

Expected result: false

### You Do pages · Independent questions

Q1. Boolean [1]; OR [1].

Q2. AND [1] — An AND condition is true only when both component conditions are true. [1]. OR [1] — An OR condition is true if at least one condition is true. [1].

Q3. Boolean [1] — A Boolean value represents true or false. [1]. OR [1] — An OR condition is true if at least one condition is true. [1]. NOT [1] — NOT reverses a Boolean value, changing true to false or false to true. [1].

### Exit Ticket · Model answer

True





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

For a alarm controller, explain why and matters. [2 marks]

Worked method: AND [1] — An AND condition is true only when both component conditions are true. [1].

### Guided 3

When designing a timetable access rule, explain how or and not should be used. [4 marks]

Worked method: OR [1] — An OR condition is true if at least one condition is true. [1]. NOT [1] — NOT reverses a Boolean value, changing true to false or false to true. [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Complete the Boolean expression so the result is false when only one check passes.

```
hasPassword = true
hasSecondFactor = false
print(hasPassword ... hasSecondFactor)
```

write

### Question 1

State two programming terms or techniques relevant to this lesson. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

### Question 2

A gym opening condition needs to work correctly. Explain how and and or would help. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

### Question 3

A pupil is programming a museum visitor rule. Explain the roles of boolean, or and not in their solution. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## Exit Ticket · Answer before checking

A Boolean value represents true or false.